

Guarded Block Transfers for Polytope Distance: Linear Convergence with Explicit Constants and Gap-Certified Screening for SVM Training

author names withheld

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Abstract

Hard-margin SVM training is the distance problem between two convex hulls, which the Triangle Algorithm (TA) solves with a certified duality gap. We propose *guarded block transfers*: k Mitchell–Demyanov–Malozemov (MDM) transfers per scan are aggregated into one exact line search, guarded by a comparison with the single best transfer at a cost independent of the number of points n , and replaced by an away step when the leading transfer is capacity-clipped. On the Minkowski-difference polytope, where the objective is 1-strongly convex, we prove global linear convergence with explicit constants and no combinatorial swap-step bounds; the per-step contraction is k -independent and only the drop-count bound grows with k , so aggregation is provably safe while the saving in scans is measured, not proved. Since the TA certificate is the Frank–Wolfe gap divided by the current distance, gap-safe screening applies, and we bound the iteration after which it has removed every zero-weight point off the optimal faces by $O\left(\frac{k}{\rho} \log \frac{1}{\tau}\right)$ up to logarithmic factors in the geometry, ρ the rate constant and τ the optimal-face margin. On five LIBSVM benchmarks the solver matches the baselines to five digits wherever both converge, ties LIBSVM on sparse-support hard margins, beats BPCG and kernel SMO on kernel L_2 margins (where LIBSVM on a precomputed kernel is faster still), and is one to two orders of magnitude slower on dense-support L_2 margins and nearly touching kernel hulls, where LIBLINEAR finishes in seconds. We report both sides.

1. Introduction

Training a linear hard-margin SVM is equivalent to finding the closest points of the two classes’ convex hulls: the optimal separator is the perpendicular bisector of the witness pair and the margin is half the hull distance [8]. Kalantari’s Triangle Algorithm (TA) exploits this *distance duality* directly [1]: TA I decides whether the hulls intersect, and TA II computes their distance while maintaining an iterate pair (p, q) and a certified interval $[LB, UB]$ from parallel supporting hyperplanes. The enhanced TA (ETA) of [2] adds dot-product caching and search heuristics.

The workhorse step of such solvers is the *weight transfer* from the worst active vertex to the best candidate, the Mitchell–Demyanov–Malozemov (MDM) step [4], which is the pairwise Frank–Wolfe (FW) step of [3] on the relevant polytope. Each step costs an $O(n)$ scan and yields one transfer; we ask how much of the scan can be reused across several transfers without weakening the guarantee.

Contributions.

1. **Guarded block transfers** (§3): pair the top- k receivers with the worst- k active donors, take one exact line search along the aggregated direction, *guard* it by an $O(k^2)$ comparison with the single MDM step, and fall back to an away step on capacity-clipped iterations.
2. **Linear convergence with explicit constants** (Theorem 3): on $Z = \text{conv}(V) - \text{conv}(W)$ the guarded block algorithm inherits the pyramidal-width rate of pairwise FW up to a constant, and the away fallback replaces the combinatorial swap-step bound by linear drop-step counting (cf. [12]).
3. **Screening activation with explicit constants** (Theorem 6): the TA certificate is the FW gap divided by the distance (Lemma 5), so gap-safe screening [17] applies, and we bound the iteration by which it has removed every zero-weight point off the optimal faces.
4. **A regime study on real data** (§5): a heuristic pyramidal-width estimate ($\sim 1/\sqrt{n}$) suggests $O(n^2)$ -type cost on dense supports, and two batteries on five LIBSVM datasets show the boundary from both sides.

Code, proofs, numerical checks and experiment scripts are in the anonymised supplementary repository.

2. Setting

Let $V = \{v_1, \dots, v_n\}$, $W = \{w_1, \dots, w_m\} \subset \mathbb{R}^d$. The polytope distance problem is

$$\min_{p \in \text{conv}(V), q \in \text{conv}(W)} \frac{1}{2} \|p - q\|^2. \quad (1)$$

Writing $z = p - q$ and $Z := \text{conv}(V) - \text{conv}(W) = \text{conv}\{v_i - w_j\}$, (1) is the minimum-norm-point problem $\min_{z \in Z} F(z)$, $F(z) = \frac{1}{2} \|z\|^2$, whose objective is 1-strongly convex and 1-smooth. Let z^* be the minimiser, $\delta^* = \|z^*\|$, $h(z) = F(z) - F(z^*)$, $M = \text{diam}(Z)$ and $\delta = \text{PW}(A) > 0$ the pyramidal width [3] of the atom set $A = \{v_i - w_j\}$ (a property of the atom set, not of Z alone).

The algorithm keeps simplex weights α on V , β on W (iterate z_t after t iterations), with $p = \sum \alpha_i v_i$, $q = \sum \beta_j w_j$; the induced product representation $\lambda_{ij} = \alpha_i \beta_j$ of z over the atoms $v_i - w_j$ has support $\text{supp } \alpha \times \text{supp } \beta$. With $g = -\nabla F(z) = q - p$, the *side scores* are $s_i^V = \langle g, v_i \rangle$ and $s_j^W = \langle p - q, w_j \rangle$, maintained incrementally from cached arrays.

Certificate. For any iterate, $\text{UB} = \|z\|$ and $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\| = (\min_i \langle z, v_i \rangle - \max_j \langle z, w_j \rangle) / \text{UB}$ satisfy $\text{LB} \leq \delta^* \leq \text{UB}$; the TA stops when the certified relative gap $(\text{UB} - \text{LB}) / \text{UB}$ is at most ε . The L_2 soft margin with regularisation parameter C , the ν -SVM and the kernel variants reduce to (1) exactly (augmented coordinates that turn the Gram matrix K into $K + I/C$; reduced convex hulls; feature-space images), see Appendix B.

Gaps. $g^{\text{FW}} = \max_{u \in Z} \langle g, u \rangle - \langle g, z \rangle$, $g^{\text{A}} = \langle g, z \rangle - \min_{u \in \text{supp } \lambda} \langle g, u \rangle$, $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}}$. Because $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, each gap splits into a V -side and a W -side term computable from the score arrays in $O(n + m)$ (Lemma 7), written $g_V^{\text{PW}}, g_W^{\text{PW}}$ (likewise for FW and A), with g_{side} the term of the side being stepped; a transfer $v_r - v_u$ on the V side is a pairwise direction of Z (Lemma 8).

Algorithm 1: One iteration of the guarded block-transfer step (side V ; W symmetric)

top- k receivers by s^V ; worst- k donors in $\text{supp } \alpha$; pair best-with-worst: (r_j, u_j) , $j \leq k'$ ($k' \leq k$ pairs, limited by the support size); pair 1 is the MDM pair (r^*, u^*)
 $d_j \leftarrow v_{r_j} - v_{u_j}$; $\text{gap}_j \leftarrow s_{r_j}^V - s_{u_j}^V$; discard pairs with $\text{gap}_j \leq 0$ and receivers that are donors (if r^* is a donor it leaves the donor list, so pair 1 survives); $\gamma_j \leftarrow \min(\text{gap}_j / \|d_j\|^2, c_j)$
 // $c_j = \alpha_{u_j}$: donor capacity
 1 **if** pair 1 uncapped ($\gamma_1 = \text{gap}_1 / \|d_1\|^2$) [case (a)] **then**
 2 $\Delta_B \leftarrow$ gain of exact line search $\theta \in [0, 1]$ along $d = \sum_j \gamma_j d_j$; $\Delta_1 \leftarrow$ gain of the MDM step
 3 take the step with the larger gain // guard, $O(k'^2)$
 4 **else**
 5 **if** the away step on u^* is boundary-clipped [case (b)] **then**
 6 take it // drop step: $\alpha_{u^*} \rightarrow 0$
 7 **else**
 8 take the best of {block, MDM, away(u^*), toward(r^*)}
 9 **end**
 10 **end**

3. Guarded block transfers

Algorithm 1 amortises one $O(n)$ scan over up to k transfers. Only the scan is amortised: a new receiver still costs one Gram or kernel column, $O(nd)$. Every candidate gain is available from cached quantities at a cost independent of n : with $S = \langle g, d \rangle = \sum_j \gamma_j \text{gap}_j$ and $D = \|d\|^2 = \gamma^\top \Gamma \gamma$, Γ the $k' \times k'$ Gram matrix of the pair directions ($O(k'^2)$ cached dot products), the block gain is $\theta S - \theta^2 D / 2$ at $\theta = \min(S/D, 1)$; the MDM, away and toward gains are the usual one-dimensional expressions. The *gain* of a step is the decrease $h(z_t) - h(z_{t+1})$. The cache update for a block is one matrix–vector product with the net per-index coefficients; columns are computed on first use and cached, so support columns cost nothing after their first appearance. Feasibility is immediate, since donors lose at most $\theta \gamma_j \leq c_j$. We call an iteration *case (a)* if pair 1 is uncapped and *case (b)* otherwise; the away step of line 6, which drives α_{u^*} to zero, is a *drop* step (a capped block pair may also zero a donor, which only helps the counts below).

Safe screening. With $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$ and $v_{\min} = \arg \min_i \langle z, v_i \rangle$, a zero-weight point with $\langle z, v_i \rangle - \langle z, v_{\min} \rangle > r \|v_i - v_{\min}\|$ is outside the optimal support and is removed (symmetrically on W with the max). Because the optimum’s support survives, the reduced problem has the same optimum and all subsequent lower bounds remain valid.

4. Analysis

Full proofs and the conventions for degenerate cases are in Appendix A.

Lemma 1 (Block gain) *Let $b_j = \gamma_j \|d_j\|$ over the k' retained pairs, all with $\text{gap}_j > 0$. Then $S \geq \sum_j b_j^2$, $D \leq k' S$, and the realised block gain satisfies $\Delta_B \geq S / (2k') \geq \Delta_1 / (2k')$, Δ_1 the MDM gain.*

With the guard, every case-(a) iteration gains $\max(\Delta_B, \Delta_1) \geq \Delta_1$: the guarded step never makes less progress than pairwise FW.

Lemma 2 (Good-step progress) *If the iteration works on the side with the larger pairwise gap, then on any non-drop iteration gain $\geq \min((g^{\text{PW}})^2/(32M^2), g^{\text{PW}}/8)$.*

Theorem 3 (Global linear convergence) *Suppose the guarded block-transfer algorithm with away fallback is run for one side per iteration (the side with the larger pairwise gap), using any $k \geq 1$, from any vertex pair (initial support $s_0 = 2$). Let $h_t = h(z_t)$. Then every non-drop iteration satisfies $h_{t+1} \leq (1 - \rho)h_t$ with*

$$\rho = \frac{1}{16} (\delta/M)^2.$$

For any horizon T , among the first T iterations the number of drop steps is at most $kT_{\text{good}} + s_0$, where T_{good} is the number of non-drop iterations; consequently, $h_T \leq h_0 (1 - \rho)^{(T-s_0)/(k+1)}$. Under the two-sided schedule of the implementation (larger-gap side first, second side skipped after a drop) every non-drop iteration has the same contraction, the drop count is at most $2kT_{\text{good}} + s_0$, and $h_T \leq h_0 (1 - \rho)^{(T-s_0)/(2k+1)}$ (Lemma 10).

The rate combines Lemma 2 with $g^{\text{PW}} \geq \delta\sqrt{2h}$ and $g^{\text{PW}} \geq g^{\text{FW}} \geq h$ (Lemma 9). Drop counting is the away-step argument: each drop removes one support index and non-drop steps add at most k . No swap-step bound is needed: a case-(b) iteration either drops (counted above) or takes the best of four candidates, of which the toward or away step alone gains Lemma 2's amount, whichever wins. Table 2 used the earlier unconditional V -then- W order (Table 3).

Proposition 4 (Near-orthogonal speed-up) *If all k' pairs are uncapped and $|\langle d_i, d_j \rangle| \leq \eta \|d_i\| \|d_j\|$ for $i \neq j$, then the block gain is at least $(\sum_j \Delta_j)/(1 + \eta(k' - 1))$, where Δ_j is pair j 's individual gain.*

As $\eta \rightarrow 0$ one scan realises the sum of all k pairwise gains.

Lemma 5 (The certified gap is the FW gap) *At any full scan, with LB evaluated at the current $z \neq 0$, $\text{UB} - \text{LB} = g^{\text{FW}}(z)/\text{UB}$ (a retained best lower bound gives $\leq$), and $g^{\text{FW}}(z) \leq 2h + R\sqrt{2h}$ with $R = \max_{u \in Z} \|u\|$, the radius of Z about the origin.*

Theorem 6 (Screening activation) *Assume $\delta^* > 0$. Let $F_V = \{i : \langle z^*, v_i \rangle = \min_j \langle z^*, v_j \rangle\}$ and, with the max, F_W be the optimal faces: the vertices of zero score margin, which contain the optimal supports (possibly strictly). Let $\tau_i = \langle z^*, v_i \rangle - \min_j \langle z^*, v_j \rangle$ be the score margin of v_i (and symmetrically on W), with $\tau > 0$ the minimal nonzero margin. Finally, let L be the larger class diameter.*

*A zero-weight point with margin τ_i is screened as soon as $r < \tau_i/(3L)$. Consequently, once $h_t < H := \tau^4 \delta^{*2}/(2592 R^4 L^4)$, every zero-weight point of V outside F_V and of W outside F_W is removed permanently at the next screening round. Under Theorem 3 this holds for every iteration $t > s_0 + \frac{a_k}{\rho} \log \max\{1, h_0/H\}$, with $a_k = k + 1$ one-sided and $2k + 1$ two-sided. Following this iteration, each subsequent scan costs $O(|F_V| + |F_W| + |\text{supp } \alpha| + |\text{supp } \beta|)$ instead of $O(n + m)$.*

In words: since $r^2 \leq 2R(\text{UB} - \text{LB})$, a point with margin τ_i is screened once the certified gap falls below $\tau_i^2/(18RL^2)$. Lemmas 1, 5 and 10, Proposition 4 and Theorem 6 are verified numerically on random and adversarial ensembles at machine precision (Appendix E).

Table 1: Regime battery on real data, five seeds, 600 s per solver, mean $\pm$ 95% CI. Kernel cells (RBF with width parameter $1/d$) use a class-balanced subsample of n_k points. Same objective per row: LIN, linear hard margin (ETA with screening vs LIBSVM’s $C=10^6$ surrogate; our SMO in brackets); KHM, feature-space hard margin (K-ETA vs the same surrogate); KL2, kernel L_2 margin, $C=1$ (K-ETA vs kernel SMO, exact objective; in brackets LIBSVM on the precomputed $K + I/C$ kernel, fit time + kernel construction). conv = seeds in which ETA certified; oracle = $O(nd)$ column (ETA) or row (SMO) evaluations, thousands. Run with the released code (fallback in every path, schedule of Lemma 10).

cell	dataset	n_k / d	ETA (s)	baseline (s)	conv	dist. ETA / base	acc. ETA / base	oracle
LIN	gisette	6000 / 5000	31.6 ± 2.8	36.5 ± 5.9 [259 $\pm$ 31]	5/5	3.6638 / 3.6640	.981 / .981	1.5 [11.1]
KHM	gisette	6000 / 5000	57.8 ± 8.7	68.1 ± 13.4	5/5	.05215 / .05215	.979 / .979	2.6
KHM	ijcnn1	9853 / 22	351 ± 109	1.1 ± 0.2	5/5	.00430 / .00430	.946 / .946	2.6
KHM	a9a	10000 / 123	> 600	19.8 ± 1.3	0/5	$\leq .00006$ / .0033	.707 / .752	5.9
KHM	covtype	10000 / 54	> 600	134 ± 3 (maxiter)	0/5	$\leq .00055$ / .00025	.694 / .712	7.5
KHM	w8a	6479 / 300	11.7 ± 32.4	—	5/5	hulls intersect	—	0.5
KL2	gisette	6000 / 5000	84.8 ± 15.4	243 ± 15 [$0.5 \pm 0.1 + 3$]	5/5	.06882 / .06882	.971 / .971	3.7 / 15.2
KL2	w8a	6479 / 300	8.7 ± 1.6	28.5 ± 1.3 [$0.6 \pm 0.1 + 1$]	5/5	.05682 / .05682	.975 / .975	3.6 / 20.2
KL2	a9a	10000 / 123	15.3 ± 2.8	27.0 ± 4.7 [$2.2 \pm 0.2 + 2$]	5/5	.03019 / .03019	.795 / .795	7.6 / 43.6
KL2	covtype	10000 / 54	21.4 ± 1.8	15.4 ± 0.6 [$2.1 \pm 0.1 + 2$]	5/5	.02609 / .02609	.792 / .792	9.1 / 56.2
KL2	ijcnn1	9853 / 22	7.7 ± 1.9	5.3 ± 0.4 [$1.1 \pm 0.1 + 2$]	5/5	.04790 / .04790	.959 / .959	4.6 / 26.1

5. Experiments

Setup. All solvers are validated against an exact QP (distances agree to 10^{-9} or better). Stopping rule $\varepsilon = 10^{-3}$ on the certified relative gap unless stated; baselines use matched tolerances on the same objective, LIBLINEAR up to its intercept penalty (Appendix C). Synthetic instances follow [2] (two Gaussian clouds, $n = 5,000$ per class). Real data are the LIBSVM sets a9a, w8a, ijcn1, gisette and covtype (100k subsample), standardised, official test sets where they exist, five seeds, 600 s per solver and cell, all on one dedicated 32-vCPU Xeon 2.6GHz machine (Appendix C). “Our SMO” is our own implementation and K-ETA the kernelised ETA.

Blocks and screening (Appendix C). Two effects separate in Table 6. The guard and away fallback alone (guarded $k=1$) cut plain pairwise FW’s iterations $1.4\text{--}2\times$ and its time $1.2\text{--}1.3\times$; aggregation cuts iterations a further $2.6\text{--}4.4\times$, but this buys wall-clock only where the $O(n)$ scan dominates the columns: $1.6\text{--}1.8\times$ on ijcn1 ($d=22$) and nothing on gisette ($d=5,000$), where guarded $k=1$ is the fastest configuration (Tables 8, 6). $k=16$ is untuned (ρ is k -independent, an oversized block costs guard work, not progress; $k=4$ and 16 agree within CI). Screening gives $1.4\text{--}2.4\times$ at $\varepsilon = 10^{-5}$ and is neutral at 10^{-3} (Theorem 6 is a sufficient condition, which these runs stop before meeting).

Real data, sparse-support regime (Table 1). TA I first certifies separability (gisette yes, the other four no, in $0.1\text{--}35$ s), a diagnostic no baseline produces. On gisette ($d=5,000$; 1,237 of 6,000 points in the support) ETA ties LIBSVM on the linear hard margin, both $7\text{--}8\times$ ahead of our SMO; K-ETA is $1.2\times$ faster than LIBSVM on the RBF hard margin and $2.9\times$ faster than kernel SMO on the kernel L_2 margin, with identical solutions. Across all five KL2 cells K-ETA matches kernel SMO’s solution with $4\text{--}6\times$ fewer kernel-column evaluations; in wall-clock it is $3\times$ faster where columns

are expensive (gisette, w8a) and $1.4\times$ slower where they are cheap (ijcnn1, covtype). Against BPCG [12] with the same column cache and certificate (Table 7), block ETA needs $12\text{--}35\times$ fewer iterations, is $2.5\text{--}5.6\times$ faster on the five KL2 cells and ties on the gisette L_2 margin, where columns dominate. LIBSVM on the precomputed $K + I/C$ kernel (brackets in Table 1) reaches the same distances $2.5\text{--}24\times$ faster than K-ETA, kernel construction included: the KL2 wins are over column-oracle solvers only.

The KHM column shows where the geometry turns against the method. On ijcnn1 both return the same solution (819 support vectors), but K-ETA needs $1.6\text{--}2.7 \times 10^5$ iterations and 351 s against LIBSVM’s 1.1 s: the hulls are 0.004 apart in a space of diameter $\approx \sqrt{2}$, so a relative certificate needs an absolute gap of order 10^{-6} (a log-factor cost; δ/M is translation-invariant and unknown here), which suits LIBSVM’s second-order selection far better. On a9a and covtype the hulls are within 10^{-4} and 5×10^{-4} of touching: K-ETA times out with an uncertified iterate and LIBSVM’s $C=10^6$ surrogate stops at its KKT tolerance (iteration cap on covtype), so neither certifies anything. On w8a the hulls intersect (duplicate points, opposite labels); K-ETA certifies this in 12 s.

Real data, L_2 soft margins (Table 2, appendix). The same boundary appears within one problem class. Gisette is separable, so its L_2 margin has a sparse support and ETA converges to the certified gap in 30–45 s at every C , $7\times$ faster than our SMO. LIBLINEAR at matched tolerance (10^{-6}) takes 8 min to 1.5 h; at its default 10^{-4} it takes 5–66 s but stops 5–24% above the certified optimum with lower test accuracy (Table 4). ETA has no wall-clock advantage over LIBLINEAR on gisette; it adds the certificate that the fast answer is not the optimum. On the four overlapping datasets both geometric solvers hit the cap on every cell while LIBLINEAR finishes in 0.3–39 s (Table 5); at $C \geq 1$ the capped iterate is not usable, on covtype at no C . The uncapped traces (Figure 1) show the linear-rate signature of Theorem 3 with 60–65% of points in the support. Heuristically (δ has no known bound in terms of the class hulls), $n_s \approx 0.6n$ active atoms forming a near-regular simplex have pyramidal width $2/\sqrt{n_s}$ times the side [3], so $\rho \sim 1/n_s$ and the cost is $O(n^2)$ -type.

6. Related work

Distance duality and the TA [1, 2]; MDM [4, 25], Gilbert [5] and Wolfe [26]; FW for polytope distance and SVMs [21–24]; the nearest-point and reduced-hull view [6, 8]; away steps [7], the pyramidal-width analysis [3] and the facial distance [27]. Boosted FW [13], which also aggregates atoms into one direction, swap-step-free pairwise variants [12, 14] and block-coordinate FW [9] are the closest algorithmic relatives; gap-safe screening [15, 17, 18] and active-set identification [19, 20] the closest to Theorem 6, with [10, 11, 16] as its SVM ancestors (Appendix D).

7. Conclusion

Guarded block transfers give the Triangle Algorithm a linear rate with explicit constants and no combinatorial swap-step bounds, and a safe screening rule with a finite activation bound. The experiments locate the boundary: sparse support and well-separated hulls, where the solver ties LIBSVM, beats the column-oracle solvers BPCG and kernel SMO (not LIBSVM with the kernel in memory), and supplies a certificate, against dense support or nearly touching hulls, where LIBLINEAR or LIBSVM is one to two orders of magnitude faster. Limitations: screening pays only when the certificate closes; the implementation is NumPy-level and dense; the constants are loose; the ν -SVM variant is not analysed; and k is fixed.

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Appendix A. Proofs

Conventions. $M > 0$ (a singleton Z is solved at initialisation), and the algorithm stops with “intersect” when $UB = \|z\| = 0$, before any division by UB . Lemma 1 and Algorithm 1 presuppose $k' \geq 1$: a side with no positive-gap pair is skipped, and on a non-drop iteration the side with the larger pairwise gap has one. An away direction whose donor has weight one is unavailable (the iterate is that vertex and the away gap of that side is zero). In Proposition 4, $\eta \geq 0$.

Lemma 7 (Separability) *With the product representation, $g^{\text{PW}} = g_V^{\text{PW}} + g_W^{\text{PW}}$ where $g_V^{\text{PW}} = \max_i s_i^V - \min_{i \in \text{supp } \alpha} s_i^V \geq 0$ and g_W^{PW} likewise; the same holds for g^{FW} and g^{A} .*

Proof. $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, so the max over atoms and the min over the support separate. Each side term is nonnegative: the FW term is the maximum score minus the weighted mean of the support scores, the away term is that mean minus the support minimum, and a weighted mean lies between the minimum and the maximum. ■

Lemma 8 (Side transfers are Z -pairwise directions) *For $\alpha_u > 0$ and any $b \in \text{supp } \beta$, $v_r - v_u = (v_r - w_b) - (v_u - w_b)$ is a difference of an atom of Z and an atom in the support, and the transfer $\gamma \leq \alpha_u$ realises $z \leftarrow z + \gamma(v_r - v_u)$ inside Z . The side capacity $\alpha_u = \sum_b \alpha_u \beta_b$ is the total weight of the product atoms $v_u - w_b$, $b \in \text{supp } \beta$: the side transfer is the sum of the atom-pair transfers of masses $\gamma \beta_b$ from $v_u - w_b$ to $v_r - w_b$, and for $r \neq u$ this weight update is feasible exactly when $0 \leq \gamma \leq \alpha_u$; hence every such transfer remains in Z .*

Proof of Lemma 1. Since $\gamma_j \leq \text{gap}_j / \|d_j\|^2$, we have $\gamma_j \text{gap}_j \geq \gamma_j^2 \|d_j\|^2 = b_j^2$. Consequently, $S = \sum_j \gamma_j \text{gap}_j \geq \sum_j b_j^2$. By the triangle inequality, $\|d\| \leq \sum_j \gamma_j \|d_j\| = \sum_j b_j$, and by Cauchy-Schwarz, $(\sum_j b_j)^2 \leq k' \sum_j b_j^2 \leq k' S$; hence $D \leq k' S$. Exact line search on the 1-smooth F gives $\Delta_B = S^2 / (2D) \geq S / (2k')$ in the interior case and $\Delta_B = S - D/2 \geq S/2$ at the boundary $\theta = 1$, where $D \leq S$. Finally, $S \geq \gamma_1 \text{gap}_1 \geq \Delta_1$, since all terms are nonnegative and $\Delta_1 = \gamma_1 \text{gap}_1 - \gamma_1^2 \|d_1\|^2 / 2$. ■

Proof of Lemma 2. Let $g_S = \max(g_V^{\text{PW}}, g_W^{\text{PW}}) \geq g^{\text{PW}} / 2$. Every direction below is a difference of two points of one class hull, so its norm is at most $\text{diam}(\text{conv } V) \leq M$. For a feasible direction with directional derivative $\sigma > 0$, squared norm D and maximal step $\bar{\theta}$, exact line search on a 1-smooth function gives gain $\geq \min(\sigma^2 / (2D), \bar{\theta} \sigma / 2)$ (interior optimum, or the clipped step evaluated with $\bar{\theta} D \leq \sigma$). In case (a), the guard's gain is at least the MDM gain, whose derivative is g_S and whose interior optimum is attained, yielding gain $\geq g_S^2 / (2M^2) \geq (g^{\text{PW}})^2 / (8M^2)$. In case (b), non-drop, $g_S = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$, so one term is at least $g_S / 2 \geq g^{\text{PW}} / 4$. If it is the FW term, the toward step (maximal step 1) gives gain $\geq \min((g_{\text{side}}^{\text{FW}})^2 / (2M^2), g_{\text{side}}^{\text{FW}} / 2) \geq \min((g^{\text{PW}})^2 / (32M^2), g^{\text{PW}} / 8)$; if it is the away term, the away line search is interior (non-drop) and gain $\geq (g_{\text{side}}^{\text{A}})^2 / (2M^2) \geq (g^{\text{PW}})^2 / (32M^2)$. The algorithm takes the best available candidate. ■

Lemma 9 (Geometric strong convexity; [3, Thm. 6] with $\mu = 1$) *For every $z \in Z$ with a representation whose support contains an away atom, $g^{\text{PW}}(z) \geq \delta \sqrt{2h(z)}$ and $g^{\text{PW}}(z) \geq g^{\text{FW}}(z) \geq h(z)$.*

Proof. This is Theorem 6 of [3] for a 1-strongly convex objective; we include the short argument. If $z = z^*$ both inequalities are immediate ($h = 0$ and all gaps are nonnegative). Otherwise let $e = (z^* - z) / \|z^* - z\|$; the pyramidal-width inequality in the proof of [3, Thm. 6] (Definition 3 there, applied to the direction g together with $\langle g, e \rangle \leq \|g\|$) gives, for the FW atom s and the away atom v , $\langle g, s - v \rangle \geq \delta \langle g, e \rangle$. By 1-strong convexity, $\langle g, z^* - z \rangle \geq h + \frac{1}{2} \|z^* - z\|^2$, so with $\varrho = \|z^* - z\|$, $g^{\text{PW}} \geq \delta(h/\varrho + \varrho/2) \geq \delta \sqrt{2h}$ by AM-GM. Finally $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}} \geq g^{\text{FW}} = \max_u \langle g, u - z \rangle \geq \langle g, z^* - z \rangle \geq h$ by convexity. ■

Proof of Theorem 3. By Lemmas 2 and 9 a non-drop iteration gains at least

$$\min(2\delta^2 h / (32M^2), h/8) = (\delta/M)^2 h / 16$$

using $\delta \leq M$; hence $h_{t+1} \leq (1 - \rho) h_t$. Drop counting: non-drop iterations add at most k support indices, drop iterations add none and remove at least one, and the support size stays ≥ 2 from s_0 ; so the number of removals, hence of drops, is at most $kT_{\text{good}} + s_0$, so that

$$T_{\text{good}} \geq (T - s_0) / (k + 1).$$

All steps, drops included, are descent steps, so h is monotone. $\blacksquare$

Lemma 10 (Two-sided schedule) *Let every iteration compute both side gaps, take the guarded step of Algorithm 1 on the side with the larger pairwise gap first, and take the other side's guarded step only if the first step was not a drop; an iteration counts as a drop iff its first step is a drop. Then every non-drop iteration gains at least $\min((g^{\text{PW}})^2/(32M^2), g^{\text{PW}}/8)$ and the number of drops is at most $2kT_{\text{good}} + s_0$; hence Theorem 3 holds for this schedule with the same ρ and exponent $(T - s_0)/(2k + 1)$.*

Proof. On a non-drop iteration the first step is exactly the step analysed in Lemma 2, so it gains at least the stated amount; the second step is an exact line search along a feasible descent direction (or is skipped), so it cannot increase h . Drop counting: a drop iteration takes a single away step that removes one support index and adds none, because the second side is skipped; a non-drop iteration takes two guarded steps and adds at most k indices each. Removals, hence drops, therefore number at most $2kT_{\text{good}} + s_0$. $\blacksquare$

Remark 11 *The skip is what makes the count work. If the second side always steps, an iteration whose analysed side drops can add up to k indices on the other side, and the count becomes $T_{\text{drop}} \leq 2kT_{\text{good}} + kT_{\text{drop}} + s_0$, which is vacuous for $k \geq 1$.*

Proof of Proposition 4. $D = \sum_{i,j} \gamma_i \gamma_j \langle d_i, d_j \rangle \leq \sum_j b_j^2 + \eta \sum_{i \neq j} b_i b_j \leq (1 + \eta(k' - 1)) \sum_j b_j^2 \leq (1 + \eta(k' - 1))S$ by Cauchy–Schwarz on the cross terms. If $D \geq S$ the line search is interior and $\Delta_B = S^2/(2D) \geq S/(2(1 + \eta(k' - 1)))$; if $D < S$ then $\theta = 1$ and $\Delta_B = S - D/2 > S/2$, which is larger. For uncapped pairs $S = \sum_j 2\Delta_j$. $\blacksquare$

Proof of Lemma 5. $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\|$ by definition of the supporting hyperplane bound and $\text{UB} = \|z\|$; subtracting, $\text{UB} - \text{LB} = (\langle z, z \rangle - \min_u \langle z, u \rangle) / \|z\| = g^{\text{FW}} / \text{UB}$. At the optimum $0 = \langle z^*, z^* \rangle - \min_u \langle z^*, u \rangle$; subtracting from $g^{\text{FW}}(z)$, $g^{\text{FW}}(z) = (\|z\|^2 - \|z^*\|^2) + (\min_u \langle z^*, u \rangle - \min_u \langle z, u \rangle) \leq 2h + \max_u \langle z^* - z, u \rangle \leq 2h + R\|z - z^*\| \leq 2h + R\sqrt{2h}$ by 1-strong convexity. $\blacksquare$

Proof of Theorem 6. Strong convexity gives $\|z - z^*\| \leq r$ for the screening radius $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$ (τ is defined only if some point has positive margin; otherwise there is nothing to screen). Let $m^* = \min_j \langle z^*, v_j \rangle$, choose $j' \in F_V$ and set $v' = v_{j'}$; since $v_{\min}$ minimises $\langle z, \cdot \rangle$, $\langle z^*, v_{\min} \rangle - m^* = \langle z, v_{\min} - v' \rangle + \langle z^* - z, v_{\min} - v' \rangle \leq rL$. Hence $\langle z, v_i - v_{\min} \rangle = \langle z^*, v_i - v_{\min} \rangle + \langle z - z^*, v_i - v_{\min} \rangle \geq \tau_i - rL - r\|v_i - v_{\min}\|$, and the rule fires when this exceeds $r\|v_i - v_{\min}\|$, for which $\tau_i > 3rL$ suffices since $\|v_i - v_{\min}\| \leq L$. For the activation time, $r^2 \leq 2R(\text{UB} - \text{LB})$ (for $\text{LB} \geq 0$, $r^2 = (\text{UB} - \text{LB})(\text{UB} + \text{LB})$ and $\text{UB} + \text{LB} \leq 2R$; for $\text{LB} < 0$, $r^2 = \text{UB}^2$ and $\text{UB} \leq 2(\text{UB} - \text{LB})$), and $2R(\text{UB} - \text{LB}) \leq 2Rg^{\text{FW}}/\text{UB} \leq (2R/\delta^*)(2h + R\sqrt{2h}) \leq (4R^2/\delta^*)\sqrt{2h}$ (Lemma 5, with equality in the first step for the freshly evaluated bound; the last step uses $h \leq R^2/2$, which holds for every feasible iterate since $h \leq \frac{1}{2}\|z\|^2$); requiring $r^2 \leq \tau^2/(9L^2)$ yields $h_t < H$, and Theorem 3 with $\log(1/(1 - \rho)) \geq \rho$ converts it into the iteration bound (the $\max\{1, \cdot\}$ covers $h_0 \leq H$, where the condition holds from the start by monotonicity). The removal is executed at the first screening round after the condition holds; permanence follows from the safety of the rule. $\blacksquare$

Appendix B. Exact reductions

L_2 soft margin. The problem $\min \frac{1}{2}\|w\|^2 + \frac{C}{2} \sum_i \xi_i^2$ is the hard-margin problem on the augmented points $\tilde{x}_i = [x_i; e_i/\sqrt{C}]$, e_i the i -th standard basis vector, whose Gram matrix is $K + I/C$ (K the Gram matrix of the original points) and whose augmented hulls never intersect; let δ_C denote their distance. The augmentation is never materialised: the extra coordinates of an iterate are exactly its convex weights, so every augmented inner product is a cached original-space product plus a sparse diagonal correction, $\langle \tilde{p}, \tilde{v}_i \rangle = \langle p, v_i \rangle + \alpha_i/C$ and $\|\tilde{p}\|^2 = \|p\|^2 + \|\alpha\|^2/C$. The separator is recovered as $w = 2(p - q)/\delta_C^2$, $b = (\|\tilde{q}\|^2 - \|\tilde{p}\|^2)/\delta_C^2$, and the primal value is $P = 2/\delta_C^2$. Validation: the solver agrees with an exact QP on the explicitly augmented problem to 10^{-10} across random instances and values of C ; the regularisation path δ_C is monotone in C ; on a9a ($C=0.1$) its certified primal 687.07 equals LIBLINEAR’s to five digits.

ν -SVM. The L_1 (hinge) soft margin is the nearest-point problem between *reduced* convex hulls $R(V, \mu) = \{\sum a_i v_i : \sum a_i = 1, 0 \leq a_i \leq \mu\}$, $\mu = 2/(\nu\ell)$ with $\ell = n + m$ the number of training points [8]. Two changes: the extreme-point oracle becomes a capped top- k blend (weight μ on the $\lfloor 1/\mu \rfloor$ best points in the search direction, via a partial sort, still $O(n)$), and the primary step is the capped MDM transfer with blended toward steps as fallback; caching and the reduced support-function lower bound carry over unchanged. Validation: agreement with the exact box-constrained QP to 10^{-9} – 10^{-14} and with LIBSVM’s NuSVC to $\sim 10^{-10}$ through the mapping $\delta^* = \|w_{\text{svc}}\|/\varsigma$, w_{svc} LIBSVM’s weight vector and ς the per-class sum of the rescaled dual coefficients. The reduced hull is a polytope, so (1) still applies, but its vertices are blends and the capped transfer is not the pairwise step analysed in §4; the analysis of this variant is left open.

Kernels. The algorithm touches data only through inner products, so kernelisation changes plumbing, not algorithm: Gram columns become kernel columns $\kappa(V, x_i)$, squared norms become kernel diagonals, iterates exist purely as convex weights (the exact cache refresh is reconstructed from the support’s cached kernel columns), and the classifier is the kernel expansion of the witness-pair bisector. A ridge $K + I/C$ gives the kernel L_2 margin by the same diagonal trick. Validation: the linear kernel reproduces the Euclidean solver to 10^{-12} ; RBF distances match the exact kernel QP to 10^{-9} and LIBSVM’s hard-margin solution, via $\|w_{\mathcal{H}}\|^2 = (\alpha \circ y)^\top K(\alpha \circ y)$, $w_{\mathcal{H}}$ the feature-space normal, for LIBSVM’s dual coefficients α and labels y , to 10^{-5} ; kernel SMO matches the QP to 10^{-8} .

Appendix C. Experimental details

Solver settings and tolerance equivalence. ETA: block transfers with $k = 16$, guarded, away fallback, pairwise anti-zig-zag, stop at certified relative gap 10^{-3} , iteration cap 2×10^6 , 600 s cap. Tables 1, 6 and 7 were produced by the released code: the schedule of Lemma 10 and the case-(b) fallback in every path. Tables 3 and 8 and Figure 3 were rerun with it. Table 2 and the traces of Table 5 come from the submitted version, which used the unconditional V -then- W order (Table 3 shows the two schedules reach the same solutions with different iteration counts) and whose single-surviving-pair path took the possibly capacity-clipped MDM transfer directly, bypassing the fallback; the fix changes a block trajectory only on iterations with a single surviving pair (the $k \geq 4$ rows of Tables 6 and 8 have identical iteration counts under both versions), and every solution of Figure 3 agrees to 10^{-4} under both. Block assembly copies the k' cached columns into a $k' \times n$ array, an

$O(k'n)$ copy, before the $O(k'^2)$ products; screening (every 25 iterations) only in the LIN cell, where it never activated at 10^{-3} (Theorem 6 gives a sufficient condition only, which these runs stop before meeting). Our SMO: maximal-violating-pair selection, KKT gap 10^{-3} , same caps; hard margin as $C = 10^{12}$; kernel SMO on $K + I/C$. LIBSVM on the precomputed $K + I/C$ kernel (scikit-learn SVC, kernel='precomputed', $C=10^6$, tolerance 10^{-3} , five seeds; src/libsvm_kl2.py) solves the same kernel L_2 margin: the RBF kernel with $1/C$ on the diagonal is built with BLAS in 1–3 s and the fit takes 0.5–2.2 s, with distances equal to K-ETA's to five digits. LIBLINEAR (scikit-learn 1.8 LinearSVC): squared hinge with parameter $C/2$, so that its objective equals $\frac{1}{2}\|w\|^2 + \frac{C}{2} \sum \xi_i^2$; tolerance 10^{-6} , chosen so that its final primal matches ETA's certified value on every cell where ETA converges except gisette at $C=10$ (a9a 687.07 vs 687.07; gisette 0.14900 vs 0.14893, inside ETA's certified 8×10^{-4} ; ijcnn1 550.6 vs 550.5 at ETA gap 0.01); this tolerance is also why LIBLINEAR needs 18,000 iterations and over an hour on gisette at $C=1$, and 35,000 iterations and 1.5 h at $C=10$, where it nevertheless stops 5% above the certified optimum (0.1566 vs 0.1490). The gisette LIBLINEAR times in Table 2 are therefore the cost of matching ETA's primal; Table 4 reports the same configuration at scikit-learn's default tolerance 10^{-4} (five seeds, run separately without concurrent load): 5–66 s, but primals 5–24% above the certified optimum and lower test accuracy. Intercept scaling 100, so LIBLINEAR regularises $b/100$ where the geometric formulation leaves b free (the primals agree to five digits); iteration cap 2×10^5 . LIBSVM (scikit-learn SVC): tolerance 10^{-3} , 1000 MB cache, iteration cap 2×10^6 (reported as *maxiter* when hit); hard margins as $C = 10^6$; RBF kernel $\exp(-\|x - x'\|^2/d)$. Oracle counts: ETA counts Gram or kernel columns evaluated against a full class ($O(nd)$ or $O(n_k d)$ each); SMO counts kernel rows, the same unit. Columns are cached without eviction and the block Gram matrix Γ is assembled from four cached entries per pair, so peak cache memory is the column count times n_k doubles: 0.73 GB for the largest cell (covtype KL2, 9,100 columns), 0.61 GB for a9a KL2 and 0.36 GB for ijcnn1 KL2; peak process memory was not recorded.

Data and seeds. a9a, w8a, ijcnn1 and gisette use the LIBSVM training and test files; covtype (binary, scaled) is subsampled to 100,000 points and split 75/25 by seed. Features are standardised on the training set. Kernel cells use a class-balanced subsample of at most 5,000 points per class (w8a's minority class has 1,479 points, hence $n_k = 6,479$; gisette uses all 6,000). Seeds vary the covtype split, the kernel subsamples and the timing environment; on datasets with fixed splits the deterministic solvers return identical solutions across seeds, which is why several accuracy CIs are zero. Confidence intervals are t -based 95% intervals over seeds.

Hardware and concurrency. One dedicated 32-vCPU Intel Xeon 2.6 GHz Lightning Studio, 122 GB RAM, Linux, Python 3.12, NumPy 2.4 with OpenBLAS 0.3.31, OMP_NUM_THREADS=1. The L_2 battery ran 14 cells concurrently (its gisette, covtype and w8a $C=10$ cells were completed in later passes with 11–14 workers and no other load); the regime battery of Table 1 ran 12 cells concurrently with the ablation of Table 6, the BPCG runs of Table 7 and the LIBSVM kernel runs, 19 processes in all, which is why Table 6's gisette $k=16$ time (37.8 s) exceeds an earlier unloaded run's (27.8 ± 4.4 s); Table 4 was made with no concurrent load; the five trials of Figure 3 ran as seven concurrent single-threaded jobs on an otherwise idle machine and reproduce the submitted run within its intervals; the absolute times of Tables 3 and 8 are about $3\times$ those of an earlier run with identical iteration counts, so their load is uncertain and only their within-table ratios should be read. The uncapped a9a/w8a runs in Figure 1 were made earlier on the same machine under a different load (LIBLINEAR took 12 and 48 s there against 4 and 23 s in the capped battery), so ratios

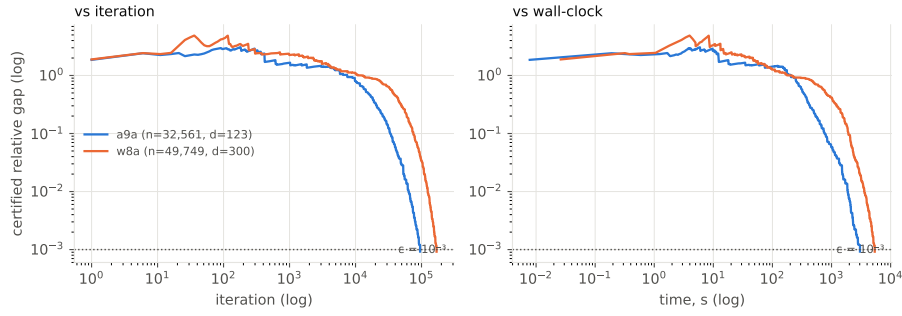
ETA on real data (L_2 soft margin, $C = 0.1$): certified gap to the 10^{-3} stopping rule

Figure 1: Certified relative gap of ETA on a9a and w8a (L_2 , $C = 0.1$, uncapped), versus iteration and wall-clock: linear-rate convergence in the dense-support regime, at $O(n^2)$ -type cost.

across the two runs carry a factor of roughly two of uncertainty. Ratios within a battery share the load but not necessarily its effect, since dense BLAS, array scans and Python set operations contend for different resources, and the intervals measure the observed variation, not this systematic part; so only ratios whose intervals separate clearly should be read as results, and an isolated sequential rerun of the cells of Table 6 and Table 7 (`src/step_profile.py`, which also records the scan, column and step time split and peak memory) is the remaining check.

Table 2: L_2 soft-margin battery: five seeds, 600 s per solver for ETA and our SMO (our own implementation; LIBSVM does not solve this objective), LIBLINEAR at tolerance 10^{-6} (Appendix C) and unbounded; mean $\pm$ 95% CI, which on the fixed-split datasets (all but covtype) measures timing noise only. “cap” = timed out in every seed; ETA’s certified relative gap is reported at its stopping point (mean [min, max]); accuracies are those of the returned iterate.

dataset (n / d)	C	ETA (s)	SMO (s)	LIBLINEAR (s)	ETA gap at stop	acc. ETA / SMO / LIBLINEAR
gisette (6,000 / 5,000)	0.1	35.3 ± 6.8	243 ± 53	500 ± 41	8×10^{-4}	.981 / .981 / .981
	1	43.3 ± 5.3	295 ± 6	3916 ± 115	8×10^{-4}	.981 / .981 / .981
	10	33.2 ± 4.6	236 ± 36	5396 ± 197	10×10^{-4}	.981 / .981 / .977
a9a (32,561 / 123)	0.1	cap	cap	3.7 ± 0.8	0.15 [0.10, 0.23]	.849 / .845 / .849
	1	cap	cap	6.8 ± 1.7	1.26 [1.25, 1.28]	.810 / .780 / .849
	10	cap	cap	2.3 ± 0.5	1.83 [1.80, 1.86]	.635 / .763 / .849
ijcnn1 (49,990 / 22)	0.1	cap	478 ± 181	0.3 ± 0.1	0.010 [0.005, 0.016]	.918 / .918 / .918
	1	cap	cap	0.6 ± 0.2	0.68 [0.61, 0.73]	.901 / .917 / .918
	10	cap	cap	0.4 ± 0.1	1.34 [1.27, 1.40]	.656 / .852 / .918
w8a (49,749 / 300)	0.1	cap	cap	22.7 ± 4.3	0.57 [0.50, 0.66]	.987 / .985 / .987
	1	cap	cap	20.6 ± 6.6	1.17 [1.14, 1.21]	.923 / .979 / .987
	10	cap	cap	13.2 ± 1.1	2.02 [1.94, 2.14]	.720 / .971 / .987
covtype (100,000 / 54)	0.1	cap	cap	19.4 ± 15.7	1.04 [1.02, 1.05]	.568 / .660 / .756
	1	cap	cap	6.1 ± 3.6	1.42 [1.23, 1.73]	.508 / .541 / .756
	10	cap	cap	5.0 ± 1.8	2.01 [1.20, 2.37]	.517 / .514 / .756

Reference solver in the KL2 cells. The hinge-loss SVC (`rbf`, $C=1$), a different objective, took 8.6 ± 2.9 (a9a), 3.4 ± 0.3 (covtype), 83.5 ± 13.1 (gisette), 1.1 ± 0.0 (ijcnn1) and 4.9 ± 1.2 s (w8a), with test accuracies 0.789, 0.783, 0.977, 0.950 and 0.975, within one point of the L_2 solutions and on either side of them.

Step ablation, blocks and shrinking (synthetic). Iteration counts at $\varepsilon = 10^{-6}$ (single instances): $d=30$, $n=200$: toward 158,941, midpoint 148,996, away 106, pairwise 181; $d=100$, $n=1,000$: toward and midpoint fail within 300,000, away 15,886, pairwise 3,921. At the working tolerance 10^{-3} the pairwise/away gain over the midpoint heuristic is only $\sim 11\%$, which is why the heuristic sufficed in [2]. Block transfers cut iterations $3\text{--}25\times$ relative to single MDM (e.g. 43,486 $\rightarrow$ 3,001 at $k=16$ on a dense-support L_2 instance at $d=1,000$) and are the only configuration that reaches the certified tolerance on MNIST odd-vs-even. Figure 2 shows the screening speed-ups and Figure 3 the consolidated comparison, five trials per class (mean $\pm$ 95% CI): on the four hard-margin classes the optimised solver is $1.1\text{--}1.2\times$ faster than LIBSVM on average, with overlapping intervals (0.63 ± 0.04 vs 0.75 ± 0.15 s at $d=1,000$; 17.9 ± 1.6 vs 20.6 ± 3.7 s at $d=10,000$; 0.72 ± 0.13 vs 0.80 ± 0.18 s at $\varepsilon=10^{-5}$, where the paper-style configuration fails to converge in every trial; RBF 2.33 ± 0.06 vs 2.50 ± 0.12 s); on the synthetic L_2 margin ($d=1,000$, $C=1$) it ties LIBLINEAR (20.7 ± 9.7 vs 19.4 ± 4.4 s); on MNIST odd-vs-even it is $2.7\times$ slower (172 ± 20 vs 63 ± 22 s; single MDM never converges) and on the ν -SVM $1.7\times$ slower (61.0 ± 5.0 vs NuSVC’s 36.5 ± 5.6 s). Distances agree within tolerance on every row. An earlier run of the same script on a shared development container had shown the optimised solver $1.6\text{--}2.5\times$ ahead on six of the seven classes; that machine’s Python-to-C timing ratio did not reproduce on dedicated hardware, and only the dedicated numbers are reported here.

Table 3: Two-sided schedules on the synthetic instances of this appendix ($n=5,000$ per class, $k=16$): the unconditional V -then- W order used for the batteries versus the schedule of Lemma 10 (larger-gap side first, second side skipped after a drop). Drop steps and skipped second-side steps are counted over the run.

instance	schedule	time (s)	iterations	drops	skips	distance
hard margin, $d=1000$, $\varepsilon=10^{-5}$	unconditional	0.8	16	0	0	55.339365
hard margin, $d=1000$, $\varepsilon=10^{-5}$	Lemma 10	0.8	16	0	0	55.339365
L_2 overlap, $d=100$, $C=1$	unconditional	25.2	12,311	785	0	0.078400
L_2 overlap, $d=100$, $C=1$	Lemma 10	23.3	10,701	599	316	0.078400
L_2 overlap, $d=1000$, $C=1$	unconditional	12.9	2,901	182	0	0.578143
L_2 overlap, $d=1000$, $C=1$	Lemma 10	13.4	3,001	180	121	0.578143

The guarded single pair already halves the iterations of plain MDM on ijcnn1 ($1.4\times$ on gisette): the drop steps it takes (on 0.2–2% of its iterations) are what plain pairwise FW replaces by capacity-clipped swaps. Blocks then cut iterations a further $2\text{--}4\times$. Away fallbacks that drop an index occur on 1–36% of block iterations, and their share grows with k : larger blocks add more indices per iteration, and the drops remove them.

BPCG here is the blended pairwise step of [12] on the atoms $v_i - w_j$ of Z with the separable LMO of Lemma 7, exact line searches and the certificate of Lemma 5; its scores are maintained from the same cached columns, so column counts are comparable. It needs $12\text{--}35\times$ more iterations

Table 4: LIBLINEAR on gisette at its default tolerance (10^{-4}) versus the matched tolerance (10^{-6}) of Table 2; five seeds. ETA’s certified primal is 0.14835 / 0.14893 / 0.14899 at $C = 0.1 / 1 / 10$, reached in 30–45 s with test accuracy 0.981.

C	tol	time (s)	iterations	primal (vs certified)	accuracy
0.1	10^{-4}	66 ± 10	363	0.15648 (+5.5%)	.978
0.1	10^{-6}	500 ± 41	2,434	0.14835 (+0.0%)	.981
1	10^{-4}	5.5 ± 0.8	10	0.18427 (+24%)	.974
1	10^{-6}	3916 ± 115	18,063	0.14900 (+0.05%)	.981
10	10^{-4}	4.8 ± 0.1	8	0.18487 (+24%)	.973
10	10^{-6}	5396 ± 197	35,411	0.15663 (+5.1%)	.977

Table 5: Time to 1% of the optimum, seed-0 traces. The ETA columns measure *dual* progress and LIBLINEAR’s column *primal* progress, so this is not a matched primal comparison (the primal of ETA’s returned (w, b) is not recorded in the traces). For ETA the certified dual value $2/\text{UB}_t^2$ increases to the optimum, so $s_t = 1 - (2/\text{UB}_t^2)/P^*$ with P^* the best known primal (the lower of LIBLINEAR’s 10^{-6} value and ETA’s certified value); “certified 1%” is the first time the certificate itself proves $1 - \text{LB}_t^2/\text{UB}_t^2 \leq 1\%$. ETA’s a9a and w8a times are from the uncapped runs of Figure 1. For LIBLINEAR the entry is the time of the coarsest tolerance whose primal is within 1% of P^* ; where only 10^{-6} has been run it is an upper bound. On the remaining cells ETA is not within 1% at the 600 s cap ($s = 30$ –99%).

dataset	C	P^*	ETA 1%	ETA 0.1%	cert. 1%	cert. 10^{-3}	LIBLINEAR 1%
gisette	0.1	0.14835	40 s	42 s	43 s	43 s	≤ 500 s (tol 10^{-6})
gisette	1	0.14893	44 s	46 s	47 s	48 s	≤ 3916 s (tol 10^{-6})
gisette	10	0.14899	34 s	36 s	37 s	37 s	none at 10^{-4} or 10^{-6}
a9a	0.1	687.07	811 s	1,413 s	2,121 s	3,030 s	4 s
w8a	0.1	108.80	2,084 s	2,960 s	4,162 s	5,386 s	23 s
ijcnn1	0.1	550.61	301 s	472 s	> 600 s	> 600 s ($s=0.05\%$)	0.3 s

than block ETA on every cell, each an $O(n + m)$ scan; the block’s fewer scans buy 2.5 – $5.6 \times$ in wall-clock where scans dominate (KL2) and nothing where columns do (gisette L_2).

Appendix D. Extended related work

Aggregated directions. Boosted FW [13] builds its direction by greedily aligning a combination of atoms with the negative gradient, at several oracle calls per iteration, and inherits the FW rate through an alignment condition; our block fixes the pair coefficients γ_j from one scan and relies on the guard, not on alignment, for its rate.

Swap steps. Blended pairwise conditional gradients [12] also remove pairwise FW’s swap-step bound with a guard, there between the pairwise step and a FW step, and [14] avoid bad steps with a short-step chain. Our guard compares an aggregated block of k transfers with the single MDM step and falls back to an away step, so the drop-step count of away-step FW applies unchanged. Block-coordinate FW [9] blocks over *coordinates* of a product domain rather than aggregating pairs on one polytope.

Table 6: Block-size ablation on real data (ETA only, three seeds, run concurrently with Table 1’s 12 workers and the runs of Table 7): the gisette L_2 margin ($C=1$, $d=5,000$) and the ijcnn1 kernel L_2 cell of Table 1 ($d=22$, $n_k=9,853$). MDM is plain pairwise FW (single transfers, no guard, no fallback); guarded $k=1$ is Algorithm 1 with a single pair (guard and case-(b) fallback active); drops = away fallbacks that removed a support index. Every cell converged to the certified gap with identical distance and accuracy per dataset.

instance	step	k	time (s)	iterations	columns	drops
gisette L_2 , $C=1$	MDM	1	36.6 ± 1.4	4,946	1,300	0
gisette L_2 , $C=1$	guarded	1	27.7 ± 4.8	3,471	1,334	82
gisette L_2 , $C=1$	block	4	34.7 ± 3.0	1,071	1,349	80
gisette L_2 , $C=1$	block	16	37.8 ± 10.3	831	1,480	301
ijcnn1 kernel L_2 , $C=1$	MDM	1	16.0 ± 4.0	12,796	4,520	0
ijcnn1 kernel L_2 , $C=1$	guarded	1	12.8 ± 0.8	6,118	4,514	14
ijcnn1 kernel L_2 , $C=1$	block	4	7.1 ± 1.5	2,358	4,522	30
ijcnn1 kernel L_2 , $C=1$	block	16	7.9 ± 3.0	1,396	4,590	98

Table 7: BPCG [12] versus block ETA ($k=16$) on the same objective with the same certificate ($\varepsilon=10^{-3}$), the same column cache and the same 600 s cap (`src/bpcg.py`); three seeds, run concurrently with Table 1. Every cell converged; distance and test accuracy agree per cell to five digits. Columns are kernel or Gram columns computed; drops count BPCG’s capacity-clipped local steps and ETA’s away drops.

cell	solver	time (s)	iterations	columns	drops
gisette L_2 , $C=1$	BPCG	34.2 ± 20.4	10,470	1,322	87
gisette L_2 , $C=1$	ETA	34.3 ± 15.0	831	1,480	301
gisette KL2	BPCG	213 ± 28	8,702	3,641	3
gisette KL2	ETA	83.9 ± 19.0	251	3,653	4
w8a KL2	BPCG	22.3 ± 5.8	12,342	3,577	12
w8a KL2	ETA	5.0 ± 2.2	603	3,591	50
a9a KL2	BPCG	65.4 ± 5.7	28,653	7,605	24
a9a KL2	ETA	11.7 ± 2.2	1,489	7,616	35
covtype KL2	BPCG	90.6 ± 11.2	38,422	9,065	15
covtype KL2	ETA	20.3 ± 1.1	3,336	9,074	26
ijcnn1 KL2	BPCG	20.2 ± 1.6	18,034	4,512	19
ijcnn1 KL2	ETA	5.9 ± 1.1	1,396	4,590	98

Screening and identification. Gap-safe rules [15, 17, 18] screen with a radius from the duality gap and identify the support in finite time; the active-set complexity of away-step FW [19] and the strict-complementarity bounds of [20] give the same identification from the algorithm’s side. Theorem 6 is the polytope-distance instance, with constants tied to the TA certificate; the SVM rules of [10, 11] and the exact data reduction of [16] are its ancestors.

Relation to other FW variants. The toward, away, pairwise and block rows of the step ablation in Appendix C are Frank–Wolfe, away-step FW, pairwise FW and our block step on the same polytope and objective; blended pairwise conditional gradients (BPCG) [12] is compared on real data in Table 7; Wolfe’s minimum-norm-point method [26] was not run. BPCG’s blending rule compares

Table 8: Block-size ablation on the synthetic instances (current code): single MDM ($k=1$) versus blocks of k transfers. Iterations fall with k ; wall-clock does not fall in proportion, because each block iteration touches more columns.

instance	k	time (s)	iterations	columns	drops
hard margin, $d=1000$, $\varepsilon=10^{-5}$	1	0.6	301	104	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	4	0.6	36	103	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	16	0.5	16	103	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	32	0.6	16	109	0
L_2 overlap, $d=100$, $C=1$	1	26.3	72,546	1,345	0
L_2 overlap, $d=100$, $C=1$	4	27.2	24,606	1,632	416
L_2 overlap, $d=100$, $C=1$	16	23.9	10,701	2,404	599
L_2 overlap, $d=100$, $C=1$	32	27.4	7,606	3,076	305
L_2 overlap, $d=1000$, $C=1$	1	24.8	43,486	1,242	0
L_2 overlap, $d=1000$, $C=1$	4	17.7	10,071	1,330	177
L_2 overlap, $d=1000$, $C=1$	16	13.5	3,001	1,348	180
L_2 overlap, $d=1000$, $C=1$	32	13.1	1,741	1,396	170

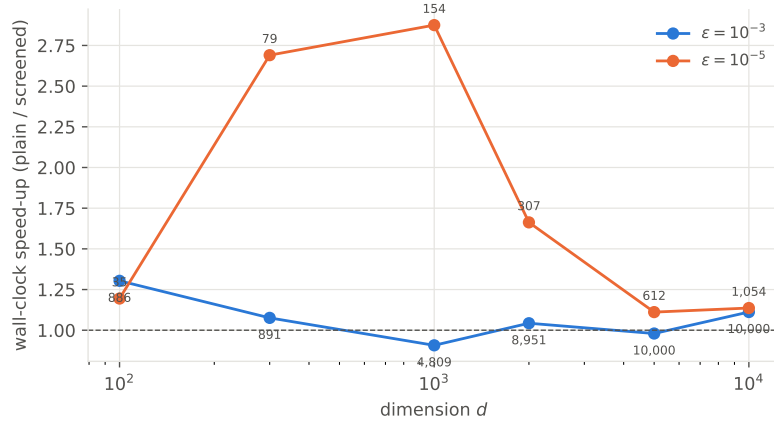


Figure 2: Safe screening: wall-clock speed-up vs dimension at two tolerances on the separated synthetic protocol; labels are surviving points of 10,000.

the away and FW gaps and its linear rate has the same pyramidal-width form as Theorem 3; our case (b) instead drops when the away step is clipped and otherwise compares the realised gains of the block, MDM, away and toward candidates, so each non-drop step is at least as good as the best single candidate, and it is the away fallback rather than the blending that yields the drop-step count. The facial distance of [27] is a computable alternative to the pyramidal width; a bound on δ for the difference atom set $A = \{v_i - w_j\}$ in terms of the two class hulls is open.

Synthetic results in brief. Away and pairwise steps converge at $\varepsilon = 10^{-6}$ where the toward and midpoint steps of [2] do not. Screening reduces 10,000 points to 36–1,054 survivors at $\varepsilon = 10^{-5}$. On seven synthetic and small-real problem classes (Figure 3, five trials each) the optimised solver is $1.1\text{--}1.2\times$ faster than LIBSVM on the four hard-margin classes (intervals overlap), ties LIBLINEAR

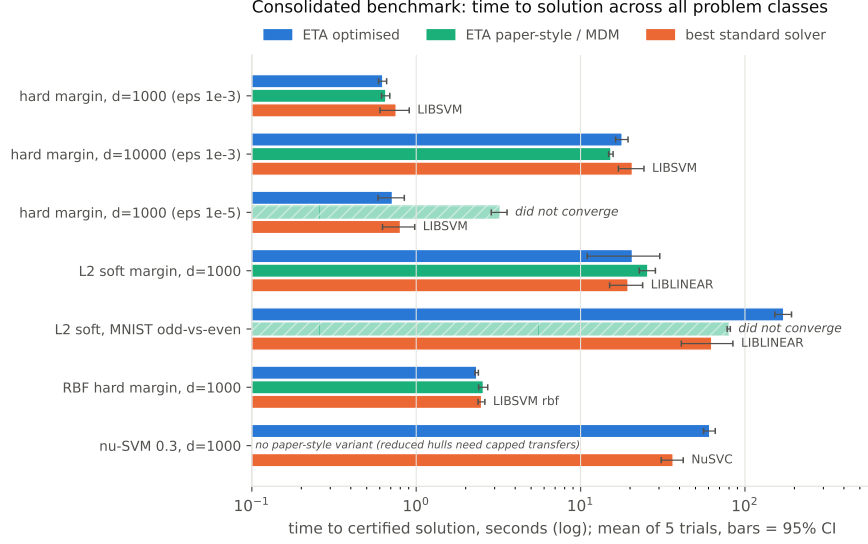


Figure 3: Time to certified solution on seven problem classes: optimised ETA vs the paper-style configuration vs the best standard solver per class; five trials per class, mean with 95% CI. The ν -SVM row has no paper-style bar because the toward-step algorithm of [2] has no reduced-hull variant (reduced hulls need capped transfers).

on the synthetic L_2 margin, and is $1.7\text{--}2.7\times$ slower than NuSVC and LIBLINEAR on the ν -SVM and MNIST classes. On the four classes where single MDM also converges, blocks and guard change wall-clock by $0.9\text{--}1.2\times$; their clear wins are the two classes (tight tolerance, MNIST) where single MDM does not converge at all.

Appendix E. Numerical verification of the analysis

Lemma 1 was checked on 4,000 random ensembles including heavily clipped and boundary cases (worst slack at machine precision); Proposition 4 on 500 near-orthogonal ensembles; Lemma 5 to 10^{-9} at arbitrary iterates; Lemma 10 on instrumented two-sided runs (no drop iteration added a support index, and every non-drop iteration met the gain bound with M replaced by the computable upper bound $\text{diam } V + \text{diam } W$); and Theorem 6 on instances whose optimal face was computed by an exact QP: at every activating round all zero-weight non-face points were screened, and the final working set equalled the optimal face exactly. Over 5,000 instrumented live block steps none increased the objective. All checks are test files in the released code.